Auckland · New Zealand

+64 27 368 7961
xiayi.shawn.001@gmail.com
linkedin.com/in/yi-xia · Google Scholar

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Yi Xia *(Sean)*

Applied AI & ML Software Engineer

Applied AI engineer who ships **production systems across computer vision and private LLMs**. Four years at **TOHU MEDIA** delivering object-detection, OCR and action-recognition pipelines, plus an AUT Research Assistant project for Tonga Police: an offline LLM application for Autopsy digital-forensics analysis. Owns the path from data and model design through Python implementation, Docker or standalone deployment, and end-user iteration. Completing an AUT PhD with two first-author Q1 journal papers, seven peer-reviewed papers and 70+ citations.

PRODUCTION Four delivered AI systems across media workflows and public-sector digital forensics.

PRIVATE AI Offline Ollama/Gemma inference · rule-plus-LLM analysis · on-device forensic data.

DEPTH PhD candidate (AUT) · first-author in two Q1 journals · seven peer-reviewed papers, 70+ citations.

TECHNICAL SKILLS

Production & Deployment

Docker · containerised model inference on cloud GPU servers · full ML lifecycle ownership · latency ↔ accuracy trade-offs

Computer Vision

Object detection (incl. YOLO family) · object tracking · OCR · action recognition · semantic / instance segmentation · classification · video analytics

ML & Programming

Python · PyTorch · TensorFlow/Keras · OpenCV · scikit-learn · NumPy · pandas · Java

LLM Systems

Ollama/Gemma · Claude/OpenAI APIs · offline inference · multi-provider integration · prompt engineering

Tooling

Git / GitHub · Linux · CUDA · Jupyter

DOMAIN EXPOSURE

- **Media / film production** - TOHU MEDIA projects: object detection, OCR & action recognition on long-form video
- **Public sector / digital forensics** - offline LLM analysis for Tonga Police Autopsy reports and risk assessment
- **Agriculture / horticulture** - published research on kiwifruit detection & tracking from orchard video (50+ citations)
- **Activity recognition** - PhD line on skeleton-based human action understanding; applications across healthcare, sports, safety and elder-care monitoring

EDUCATION

PhD, Computer & Information Sciences
Auckland University of Technology
Oct 2023 - Oct 2026 (expected)
Deep learning for skeleton-based human activity recognition.

Master of Computer & Information Sciences
Auckland University of Technology
2021 - 2023 · Specialisation: Computer Vision

BEng, Logistics Engineering (Software)
Dalian Neusoft University of Information, China
2011 - 2015

SELECTED EXPERIENCE

Artificial Intelligence Developer *at TOHU MEDIA* Auckland · Feb 2022 - May 2026

Full-time June 2025 - May 2026; previously part-time alongside MSc and PhD

- **Own the company's AI capability end-to-end**; designed and shipped **three production systems** for object detection, OCR and action recognition on client video footage (specific deliverables under NDA).
- Engineered for variable cinematography, dense overlay text, low-contrast archives, long-form takes and workloads beyond a single workstation GPU.
- Owned dataset scope, PyTorch training/evaluation, **Docker deployment to company cloud GPUs**, and iteration with editors on useful outputs, latency/accuracy trade-offs and data sovereignty.

Research Assistant *at Auckland University of Technology* Oct 2025 - Feb 2026

LLM Digital Forensics · Client: Tonga Police

- Built and delivered **Autopsy AI**, a Windows application that parses Autopsy HTML case reports, combines rule-based risk scoring with LLM analysis, and generates investigator-ready reports.
- Packaged a **100% offline Ollama/Gemma deployment** with English/Tongan risk keywords, multi-provider LLM support and a standalone executable, keeping sensitive forensic data on-device.

PhD Researcher *at Institute of Robotics & Vision, AUT* Oct 2023 - Oct 2026

Supervisors: Dr Raymond Lutui (primary) · Assoc. Prof. Sira Yongchareon

- Designed a **frequency-aware spatio-temporal graph network** for skeleton-based HAR; published in *Pattern Recognition* (Q1, IF 7.6, 2026).
- First-authored a generalisation survey in **Computer Science Review** (Q1 review, IF 17.5, 2026) and a model paper at ICONIP 2024; owned design, GPU experiments, writing and rebuttal.

Research Assistant *at School of Engineering, Computer & Mathematical Sciences, AUT* Nov 2022 - Jul 2023

- Co-built KiwiDetector and KiwiTracker for real-time orchard video; co-authored three papers and the lead paper has **50+ citations**.

Teaching Assistant *at AUT* Mar 2025 - present

- TA on multiple undergraduate computer-science courses (programming, computer vision, AI). Deliver classes and tutorials, prepare teaching materials and mark assessment.

Earlier Experience 2015 - 2023

- PB Tech, Auckland (2023): hardware diagnostics, customer service and custom PC builds. Junior Lecturer / Co-Founder, China (2015-2021): taught Java, IoT and computing; built curriculum and ran academic operations.

SELECTED PUBLICATIONS

7 peer-reviewed papers · 70+ citations

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." *Computer Science Review*. Q1 review, IF 17.5
doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." *Pattern Recognition*. Q1 journal
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." *IVCNZ 2022*, Springer LNCS. 50 citations
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14

+ three further peer-reviewed papers on kiwifruit detection & tracking - full list on Google Scholar.